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Genome assembly GRCh38.p14 reference

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NCBI RefSeq assembly

GCF_000001405.40 ⁽⁺⁾

Submitted GenBank assembly

GCA_000001405.29



Taxon

[Homo sapiens](#) (human)

Synonym

hg38

Assembly type

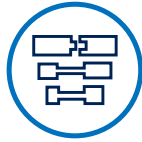
haploid with alt loci

Submitter

Genome Reference Consortium

Date

Feb 3, 2022



View annotated genes



See in Genome Data Viewer



BLAST the reference genome



Compare genomes

Assembly statistics

	RefSeq	GenBank
Genome size	3.1 Gb	3.1 Gb
Total ungapped length	2.9 Gb	2.9 Gb
Gaps between scaffolds	349	349
Number of chromosomes	24	24
Number of organelles	1	1
Number of scaffolds	470	473
Scaffold N50	67.8 Mb	67.8 Mb
Scaffold L50	16	16
Number of contigs	996	999
Contig N50	57.9 Mb	57.9 Mb
Contig L50	18	18

GC percent	41	41
Assembly level	Chromosome	Chromosome
View sequences	view RefSeq sequences	view GenBank sequences

Assembly methods

Comment

Assembly Name: GRCh38.p14 Primary Assembly

The DNA sequence is composed of genomic sequence, primarily finished clones that were sequenced as part of the Human Genome Project. PCR products and WGS shotgun sequence have been added where necessary to fill gaps or correct errors. All such additions are manually curated by GRC staff. For more information see: <https://genomereference.org>

Additional genomes

[Browse all Homo sapiens genomes \(2564\)](#)

BioProject

[PRJNA31257](#)

The Human Genome Project, currently maintained by the Genome Reference Consortium (GRC)

Publications

Genome Biol 2008

[Finishing the finished human chromosome 22 sequence](#)

Cole, Charlotte G, et al.

Nature 2006

[The DNA sequence and biological annotation of human chromosome 1](#)

Gregory, S G, et al.

Nature 2006

The DNA sequence, annotation and analysis of human chromosome 3

Muzny, Donna M, et al.

Nature 2006

DNA sequence of human chromosome 17 and analysis of rearrangement in the human lineage

Zody, Michael C, et al.

Nature 2006

Analysis of the DNA sequence and duplication history of human chromosome 15

Zody, Michael C, et al.

View in PubMed

Annotation details

See full annotation report

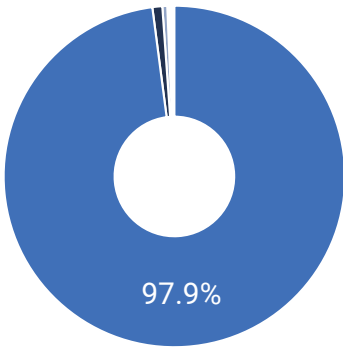
	RefSeq
Provider	NCBI RefSeq
Name	GCF_000001405.40-RS_2025_08
Date	Aug 1, 2025
Genes	59.792
Protein-coding	20.076
Pipeline	NCBI Eukaryotic Genome Annotation Pipeline (EGAP)
Software version	10.4

View RefSeq annotation

Quality analysis

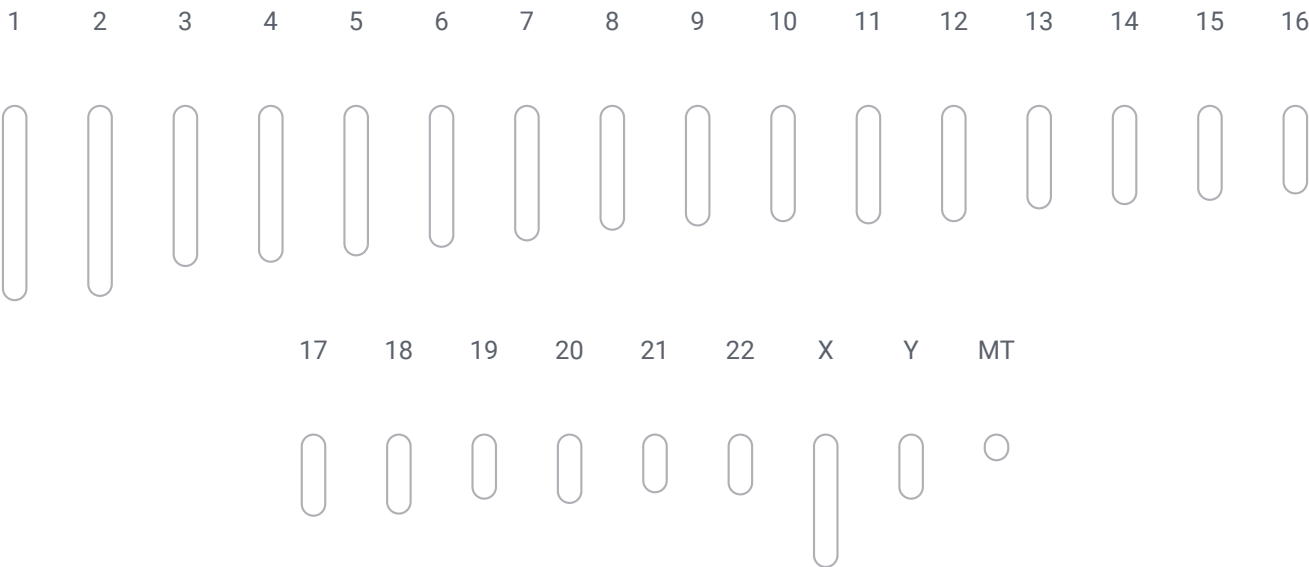
BUSCO analysis (5.7.1)

- Single_copy 97.9%
- Duplicated 0.9%
- Fragmented 0.5%
- Missing 0.7%



C:98.9%[S:97.9%,D:0.9%],F:0.5%,M:0.7%,n:13780
primates_odb10 (13780)

Chromosomes



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Chromoso...	GenBank	RefSeq	Size (bp)	GC content (%)	Action
1	CM000663.2	NC_000001.11	248.956.422	41,5	

2	CM000664.2	NC_000002.12	242.193.529	40,5	⋮
3	CM000665.2	NC_000003.12	198.295.559	40	⋮
4	CM000666.2	NC_000004.12	190.214.555	38,5	⋮
5	CM000667.2	NC_000005.10	181.538.259	39,5	⋮
6	CM000668.2	NC_000006.12	170.805.979	39,5	⋮
7	CM000669.2	NC_000007.14	159.345.973	41	⋮
8	CM000670.2	NC_000008.11	145.138.636	40	⋮
9	CM000671.2	NC_000009.12	138.394.717	41,5	⋮
10	CM000672.2	NC_000010.11	133.797.422	41,5	⋮
11	CM000673.2	NC_000011.10	135.086.622	41,5	⋮
12	CM000674.2	NC_000012.12	133.275.309	41	⋮
13	CM000675.2	NC_000013.11	114.364.328	38,5	⋮
14	CM000676.2	NC_000014.9	107.043.718	41	⋮
15	CM000677.2	NC_000015.10	101.991.189	42	⋮
16	CM000678.2	NC_000016.10	90.338.345	44,5	⋮
17	CM000679.2	NC_000017.11	83.257.441	45,5	⋮
18	CM000680.2	NC_000018.10	80.373.285	39,5	⋮
19	CM000681.2	NC_000019.10	58.617.616	48	⋮
20	CM000682.2	NC_000020.11	64.444.167	44	⋮
21	CM000683.2	NC_000021.9	46.709.983	41	⋮
22	CM000684.2	NC_000022.11	50.818.468	47	⋮

X	CM000685.2	NC_000023.11	156.040.895	39,5	
Y	CM000686.2	NC_000024.10	57.227.415	40	
MT	J01415.2	NC_012920.1	16.569	44,5	

Note: This genome assembly includes 127 unplaced scaffolds. [i](#)

RefSeq and GenBank assembly differences

The NCBI RefSeq assembly differs from the submitted assembly in the following ways:
Removed 4 unlocalized and unplaced scaffolds; RefSeq dropped two scaffolds that are predominantly rodent or bacterial in origin (KI270752.1/NT_187507.1 and KI270825.1/NT_187580.1), and dropped two unlocalized scaffolds that are now thought to be redundant with assembled chromosome sequence (KI270721.1/NT_187376.1 and KI270734.1/NT_187389.1)

Revision history

GenBank	RefSeq	Name	Level	Action
GCA_000001405.29	GCF_000001405.40	GRCh38.p14	Chromosome	
GCA_000001405.28	GCF_000001405.39	GRCh38.p13	Chromosome	
GCA_000001405.27	GCF_000001405.38	GRCh38.p12	Chromosome	
GCA_000001405.26	GCF_000001405.37	GRCh38.p11	Chromosome	
GCA_000001405.25	GCF_000001405.36	GRCh38.p10	Chromosome	
GCA_000001405.24	GCF_000001405.35	GRCh38.p9	Chromosome	
GCA_000001405.23	GCF_000001405.34	GRCh38.p8	Chromosome	
GCA_000001405.22	GCF_000001405.33	GRCh38.p7	Chromosome	

GCA_000001405.21	GCF_000001405.32	GRCh38.p6	Chromosome	⋮
GCA_000001405.20	GCF_000001405.31	GRCh38.p5	Chromosome	⋮
GCA_000001405.19	GCF_000001405.30	GRCh38.p4	Chromosome	⋮
GCA_000001405.18	GCF_000001405.29	GRCh38.p3	Chromosome	⋮
GCA_000001405.17	GCF_000001405.28	GRCh38.p2	Chromosome	⋮
GCA_000001405.16	GCF_000001405.27	GRCh38.p1	Chromosome	⋮
GCA_000001405.15	GCF_000001405.26	GRCh38	Chromosome	⋮
GCA_000001405.14	GCF_000001405.25	GRCh37.p13	Chromosome	⋮
GCA_000001405.13	GCF_000001405.24	GRCh37.p12	Chromosome	⋮
GCA_000001405.12	GCF_000001405.23	GRCh37.p11	Chromosome	⋮
GCA_000001405.11	GCF_000001405.22	GRCh37.p10	Chromosome	⋮
GCA_000001405.10	GCF_000001405.21	GRCh37.p9	Chromosome	⋮
GCA_000001405.6	GCF_000001405.17	GRCh37.p5	Chromosome	⋮
GCA_000001405.3	GCF_000001405.14	GRCh37.p2	Chromosome	⋮
GCA_000001405.1	GCF_000001405.13	GRCh37	Chromosome	⋮
n/a	GCF_000001405.12	NCBI36	Chromosome	⋮
n/a	GCF_000001405.11	NCBI35	Chromosome	⋮
n/a	GCF_000001405.10	NCBI34	Chromosome	⋮
n/a	GCF_000001405.9	NCBI34	Chromosome	⋮
n/a	GCF_000001405.8	NCBI33	Chromosome	⋮
GCA_000001405.9	n/a	GRCh37.p8	Chromosome	⋮

GCA_000001405.8	n/a	GRCh37.p7	Chromosome	⋮
GCA_000001405.7	n/a	GRCh37.p6	Chromosome	⋮
GCA_000001405.5	n/a	GRCh37.p4	Chromosome	⋮
GCA_000001405.4	n/a	GRCh37.p3	Chromosome	⋮
GCA_000001405.2	n/a	GRCh37.p1	Chromosome	⋮

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